pfi

Python for Fixed Income

COURSE 9 · DEPLOYMENT

What the Server Installs

WEEK 1 · DAY 4

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Today: what the server installs

Today

4.1 What the course environment holds

4.2 What a file needs: requirements_for

4.3 What a list costs: installed_size, Windows and Linux

4.4 The limits, as read today

4.5 The file the server reads: pins_check, not pip freeze

4.6 Four Pythons

4.7 The desk's decision

The question

What does this code import, and what does that cost to install?

A server has none of your laptop: it installs its requirements file, on a fresh Linux machine, every build.

Offline today: every size is read from files Python already has on disk.

The rule for servers

A server installs only what its code imports.

Vercel	one bundle, built from requirements.txt
A Space	Gradio preinstalled; the rest from requirements.txt
GitHub Actions	a fresh machine: installs on every run

On a platform's figures

Allowed names

included in Hobby as read

the free tier as read

the platform line

Never called

guaranteed

unlimited

free forever

What the course environment holds

189

distributions installed

3,125,180,420

bytes on disk, by RECORD

2.91 GB

1024^3 bytes, Windows

RECORD

Every distribution lists each file it installed, with its size in bytes. Add them up.

```
sum(f.size or 0 for f in d.files or [])
```

None of it goes to a server: shipping your environment sends TensorFlow to answer a DV01 request.

What a file needs: requirements_for

Import name, distribution

`import sklearn`	`scikit-learn`
`import yaml`	`PyYAML`
`import dotenv`	`python-dotenv`
`from PIL import`	`pillow`

Three servers

bondmath.py	nothing: the standard library
the bond service	`fastapi==0.143.0`
the desk, seven files	20 distributions

An upper bound: it counts imports inside functions a server may never run. Read it, keep what runs.

What a list costs, Windows and Linux

Server	Requirements	Windows	Linux
the bond service	1 line	11 distributions, 9.6 MB	9.2 MB
the desk	20 lines	94 distributions, 923.4 MB	109 wheels, 5,716.5 MB

MB of 1024 x 1024 bytes. Linux: the course build's measurement of 2026-10-12, wheels downloaded, never installed.

On Linux, torch from PyPI is the CUDA build: NVIDIA's GPU libraries come as separate wheels.

A wheel is built for one platform: measure a server's size on the server's platform.

The limits, as read today

Vercel

500 MB uncompressed bundle

A Space

Gradio from sdk_version

GitHub Actions

installs on every run

Vercel Python: a 500 MB uncompressed bundle limit (vercel.com/docs/functions/runtimes/python, read 2026-10-12).

Spaces: Gradio from sdk_version, the rest from requirements.txt (huggingface.co/docs/hub/spaces-dependencies, read 2026-10-12).

GitHub Actions: free on public repositories (docs.github.com/en/billing/concepts/product-billing/github-actions, read 2026-10-12).

The bond service fits with room to spare; the desk on Linux does not come close.

pins_check, and why not pip freeze

unpinned

the newest, whenever it builds

Windows-only

pywin32 with no marker

listed twice

FastAPI and fastapi

mcp's own requirement: a marker

```
["pywin32>=311; sys_platform == 'win32'"]
```

pip freeze: 188 lines, all pinned; pins_check:

```
line 132: pywin32 installs only on Windows; add ; sys_platform == "win32"
```

Write a server's file from the code, read it, check it. Never pip freeze.

Four Pythons

Where	Python	Set by
your laptop	3.13.8	the course's .venv
GitHub Actions	3.12 and 3.13	the workflow's matrix
Vercel	3.13	.python-version
a ZeroGPU Space	3.12.12	python_version in the README

```
Python 3.10.13: the bond service 0 excluded; the desk 10 excluded
Python 3.12.12: the bond service 0 excluded; the desk 0 excluded
```

Spaces: python_version defaults to 3.10 (huggingface.co/docs/hub/spaces-config-reference, read 2026-10-12).

State the Python everywhere, quoted in YAML.

The desk's decision

```
                        server  requirements  linux_mb against 500 MB
             the bond service             1       9.2           fits
        the desk as a snapshot             1       9.2           fits
the desk replayed on the server            20    5716.5 over, 11 times
```

Vercel Python: a 500 MB uncompressed bundle limit (vercel.com/docs/functions/runtimes/python, read 2026-10-12).

Week 2 serves a snapshot: the brief as of 2026-10-02, every figure traced to a saved exchange by its hash.

Where it fails: requirements_for reads imports, not behaviour. A file loaded at run time or a plugin found by name is not an import. The first build is the real test.

Excel and Git

```
E1:  =SUM(C2:C200)/1024^2
E2:  =IF(E1>500,"over","fits")
```

the bond service: 9.6 MB, fits

the desk: 923.4 MB, over

Excel and pandas: sizes_service.csv: 11 rows, 9.6 MB, fits | sizes_desk.csv: 94 rows, 923.4 MB, over

```
requirements-server.txt: only what the server runs
requirements-server.txt for the bond service fold...
deploykit day 4: requirements_for, installed_size...
Start of day 4: deploykit through day 3, 13 tests
```

```
 fastapi==0.143.0
-packaging==26.3
-pandas==3.0.6
-PyYAML==6.0.3
```

17 passed. requirements-server.txt keeps only what the server runs: one line.

The AI check

The docstring is the prompt. The deploy code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the lines name the import (sklearn, yaml, dotenv, PIL), not the distribution pip installs.

Code that runs is not the same as code that is right.

What you can do now

- Measure an environment from RECORD
- requirements_for: an upper bound, read it
- installed_size: Windows is not Linux
- pins_check, never pip freeze
- Four Pythons; the desk ships as a snapshot

Week 1, Day 5 Case study: bondmath through CI